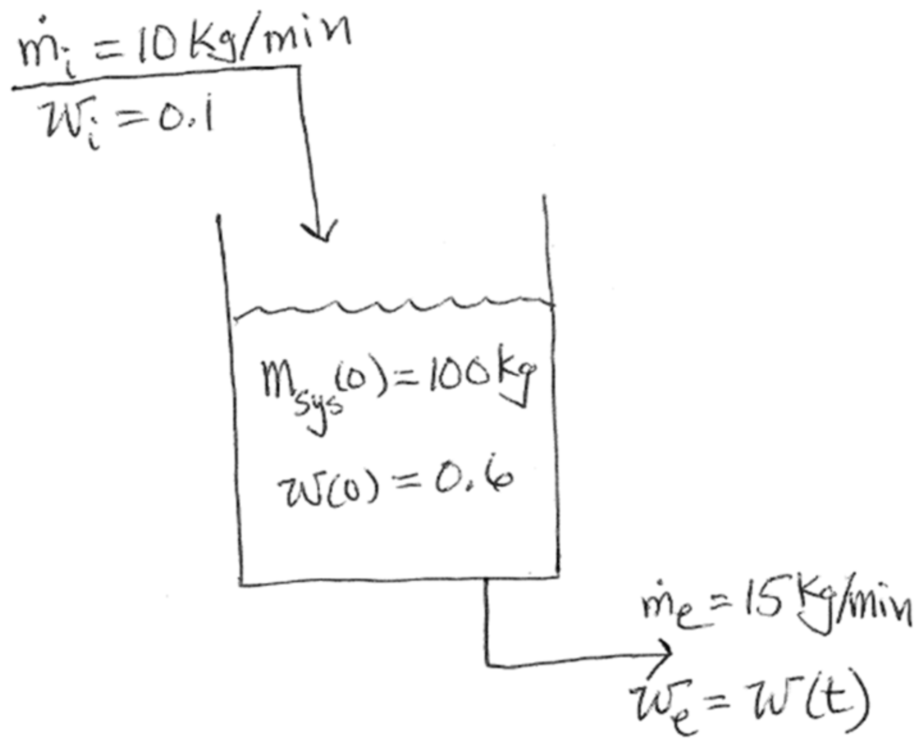
NAME ________________________________________

ChE 210 Quiz #3 13 March 2020

Consider a binary liquid mixture of species A and B. A storage tank contains 100 kg of a 60 wt% of A in the mixture. That is, the weight fraction of the A in the tank is initially $w = 0.6$. Starting at time $t = 0$, a 10 wt% of A, $w_i = 0.1$, is added at a constant rate of $\dot{m}_i$ =10 kg/min while at the same time $\dot{m}_e$ =15 kg/min of the tank contents is removed at a constant rate. Assume that the tank is well mixed so that the A weight fraction at the exit is the same as the A weight fraction in the tank, $w_e = w(t)$.

(a) Derive an expression for the total mass of the mixture $m_{sys}(t)$ in the tank as a function of time. (2 pts.)
(b) What is the total mass of the mixture in the tank in kg after 10 minutes? (1 pt.)
(c) Using the result from part (a), write a species mass balance on species A in terms of $w(t)$. Calculate the weight fraction of A in the tank after the same 10 minutes of operation $w(t = 10\ min)$ as in part (b). (7 pts.)




$$\int x^n dx = \frac{x^{n+1}}{n+1}$$

$$\int \frac{1}{x} dx = \ln|x|$$

$$\int (a+bx)\,dx = \frac{1}{2b}(a+bx)^2$$

$$\int \frac{dx}{a+bx} = \frac{1}{b}\ln|a+bx|$$

$$\int \frac{dx}{x(a+bx)} = -\frac{1}{a}\ln\left|\frac{a+bx}{x}\right|$$

$$\int \frac{dx}{(a+bx)^2} = \frac{-1}{b(a+bx)}$$

$$\int \frac{dx}{(a+fx)(c+gx)} = -\frac{1}{ag-cf}\ln\left|\frac{c+gx}{a+fx}\right|$$